

Pharmacology of Antiretroviral Agents, Part 1

Overview of antiretroviral therapy, NRTIs and INSTIs

Leslie Goldstein, Pharm.D.

Associate Professor

Department of Clinical Specialties

Division of Pharmacology

lgolds01@nyit.edu



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I am available to groups and individuals for pharmacology help and discussions by appointment.

lgolds01@nyit.edu

Following completion of the preparation materials, students should be able to:

1. Compare and contrast the within-class and between-class pharmacokinetic properties of the antiretroviral agents.
2. Describe the structure-activity relationships, mechanism of action, and mechanisms of viral resistance of each class of antiretroviral agents in relation to specifically targeted viral proteins and give examples.
3. Explain the clinical rationales for selection of specific antiretroviral therapy (ART) and give examples for:

ART-naïve patients	Transgender patients
Switch to 2-drug ART	Women (assigned female at birth)
Treatment-experienced patients	Pregnant people and peripartum care
Pre-exposure prophylaxis (PrEP)	Neonates
Post-exposure prophylaxis (PEP)	Pediatric patients
	Older patients
4. Differentiate the class and drug-specific adverse effects and their impact on patient adherence to therapy.
5. Explain the mechanism-related and metabolism- and transport-related drug interactions for specific drugs, including the antiretroviral drugs that should not be used in combination.

Links

ScholarRx Brick: Pharmacology of Antiretroviral Drugs > General Microbiology

Bricks: Antimicrobial Agents > Antiviral Drugs > Drugs to Treat and Prevent HIV Infection

<https://exchange.scholarrx.com/brick/drugs-to-treat-and-prevent-hiv-infection>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024, Chapter 49: Antiviral Agents > Antiretroviral Agents

<https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3382§ionid=281755260>

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 34: Antiviral Drugs > Section VII. Treatment of HIV Infection

<https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253329261&bookid=3222#253329364>

HIV Basic Statistics

- In 2021, 36,136 people received an HIV diagnosis in the United States and dependent areas. The annual number of new diagnoses decreased 7% from 2017 to 2021.
- An estimated 1.2 million people in the United States had HIV at the end of 2021, the most recent year for which this information is available. Of those people, about 87% knew they had HIV.
- In 2021, male-to-male sexual contact accounted for 67% (24,107) of all new HIV diagnoses in the United States and dependent areas. In the same year, heterosexual contact accounted for 22% (8,059) of all HIV diagnoses.

What you need to know and understand

- HIV life cycle → Drug targets

(1) Entry (binding and fusion to host cell), (2) reverse transcription, (3) integration of HIV cDNA into host genome, (4) replication (transcription and translation), (5) assembly and budding (6) maturation

- ART resistance: HIV reverse transcriptase is error prone and lacks a proofreading function. Mutation is frequent. Drug therapy provides the necessary *selective pressure* to promote growth of drug-resistant viruses that arise naturally. Resistance testing is important to help guide treatment decisions in both treatment-naïve and treatment-experienced patients.
- Goal of therapy: To restore immunocompetence – increase the CD4 cell count and reduce viral load to undetectable (suppression of HIV RNA replication), reduce HIV-associated morbidity and prolong the duration and quality of survival, and prevent HIV transmission
- Drug classes: nucleoside reverse transcriptase inhibitors (NRTIs), integrase strand transfer inhibitors (INSTIs), nonnucleoside/tide reverse transcriptase inhibitors (NNRTIs), protease inhibitors (PIs), entry inhibitors, and capsid inhibitor.
- Pharmacokinetic enhancers: Two pharmacokinetic enhancers (“boosters”), ritonavir and cobicistat, are potent inhibitors of CYP3A4, which increases systemic levels of the partner drug allowing for less frequent dosing and less variation in drug levels, resulting in enhanced efficacy and reduced resistance.

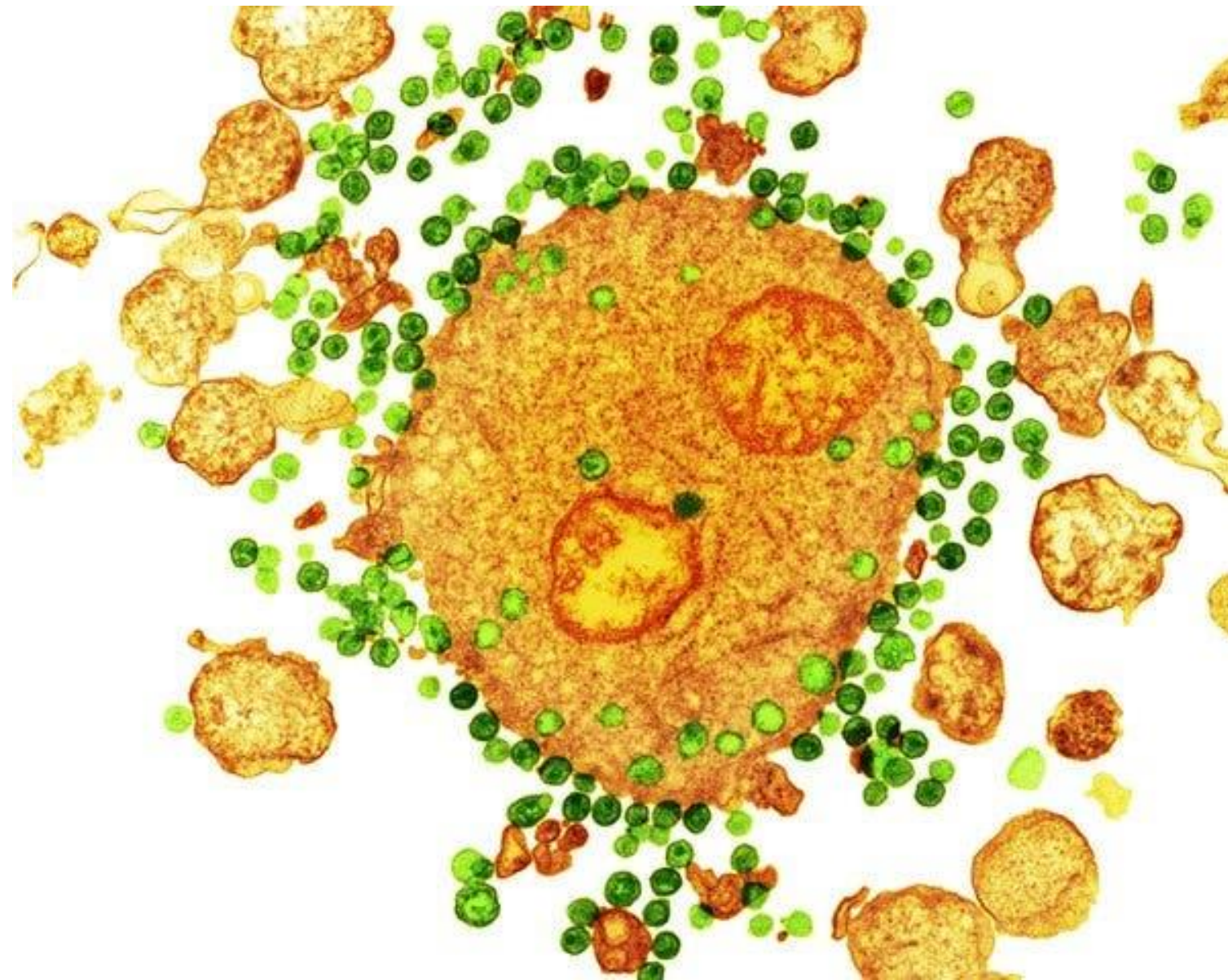
- ART principles apply across all populations.
- Recommended initial therapy in treatment-naïve patients consists of a 2 NRTI backbone with an integrase inhibitor (INSTI-based ART). An NNRTI or boosted PI backbone may be considered in certain clinical situations.
- Note: Several 2-drug combination regimens are FDA approved for stable patients who meet specific criteria.
- Treatment-experienced patients require a new regimen based on evaluation of the patient's past ART and previous and current genotype and/or phenotype of the virus. Drug resistance tests determine which drugs are likely to be fully or partially active.
- NRTIs
 - First-generation – didanosine, stavudine, zidovudine: Most toxicities are due to inhibition of mitochondrial DNA polymerase causing peripheral neuropathy, pancreatitis, and lipoatrophy and life-threatening hepatotoxicity characterized by lactic acidosis, hepatomegaly, and steatosis. Of the first-generation drugs, only zidovudine is available in the US.
 - Second-generation – tenofovir (TDF and TAF), emtricitabine, lamivudine, and abacavir. TDF can cause proximal renal tubulopathy and bone loss but TAF has less long-term kidney and bone toxicity. Lamivudine and emtricitabine are structurally similar and should not be used together. Abacavir can cause a life-threatening hypersensitivity reaction. Patients should be screened for HLA-B*5701 allele.

What you need to know and understand – Adverse Effects

- INSTIs:
 - Integrase inhibitors are bictegravir, dolutegravir, elvitegravir, and raltegravir. They are well tolerated, highly effective, and have a high barrier to resistance. They are recommended first line drugs with 2NRTIs for initial ART.
- NNRTIs:
 - This group is associated with varying degrees of GI intolerance and skin rash.
 - The two most frequently used agents, efavirenz and rilpivirine, cause QT interval prolongation and are associated with neurological and psychiatric side effects.
 - NNRTIs are metabolized by, and are inhibitors and inducers of, various CYPs. There is a high potential for drug-drug interactions.
- PIs:
 - Class side effects include insulin resistance, hyperglycemia, diabetes, hyperlipidemia, lipodystrophy, hepatotoxicity, bleeding in patients with hemophilia, and PR interval prolongation. PIs are CYP3A4 substrates and are used with boosters – CYP3A4 inhibitors. There is a high potential for drug-drug interactions.

What you need to know and understand – adverse effects

- **EIs: Entry inhibitors have different mechanisms of action, pharmacokinetic profiles, clinical uses, and adverse effects.**
 - **CCR5 antagonists:** Maraviroc is generally well tolerated. The drug has been associated with severe hepatotoxicity and related rash or fever. Liver function monitoring is recommended.
 - **Fusion inhibitor:** Enfuvirtide is a peptide given by subcutaneous injection twice daily and often causes local cutaneous reactions.
 - **Attachment inhibitor:** Fostemsavir is an oral prodrug (active temsavir). Its most common side effect is nausea. Other effects are QT interval prolongation and elevated liver enzymes in patients with hepatitis B or hepatitis C viral infections. Strong CYP3A4 inducers should be avoided.
 - **Post attachment inhibitor:** Ibalizumab is given by intravenous infusion every 2 weeks. The most common adverse effects are diarrhea, dizziness, nausea, and rash.
- **Capsid inhibitor:** Lenacapavir directly binds the interface between capsid protein (p24) subunits in hexamers, changing its conformation, which stabilizes it, prevents viral uncoating, and disrupts the viral lifecycle at several steps.



A colored transmission electron micrograph of the H.I.V. virus, in green, attaching to a white blood cell, in orange.
Credit...NIBSC/Science Source

https://www.nytimes.com/2019/12/23/health/what-we-learned-2019.html?te=1&nl=evening-briefing&emc=edit_ne_20191228

<https://clinicalinfo.hiv.gov/en> is an excellent resource.

It offers information on HIV/AIDS treatment, prevention, and research.

“Care for HIV/AIDS patients has shifted from specialists to primary care physicians.”¹

- Infectious diseases experts encourage doctors to participate in training opportunities and form relationships with local HIV/AIDS specialists to ensure they have a basic understanding of the disease.^{1,2}

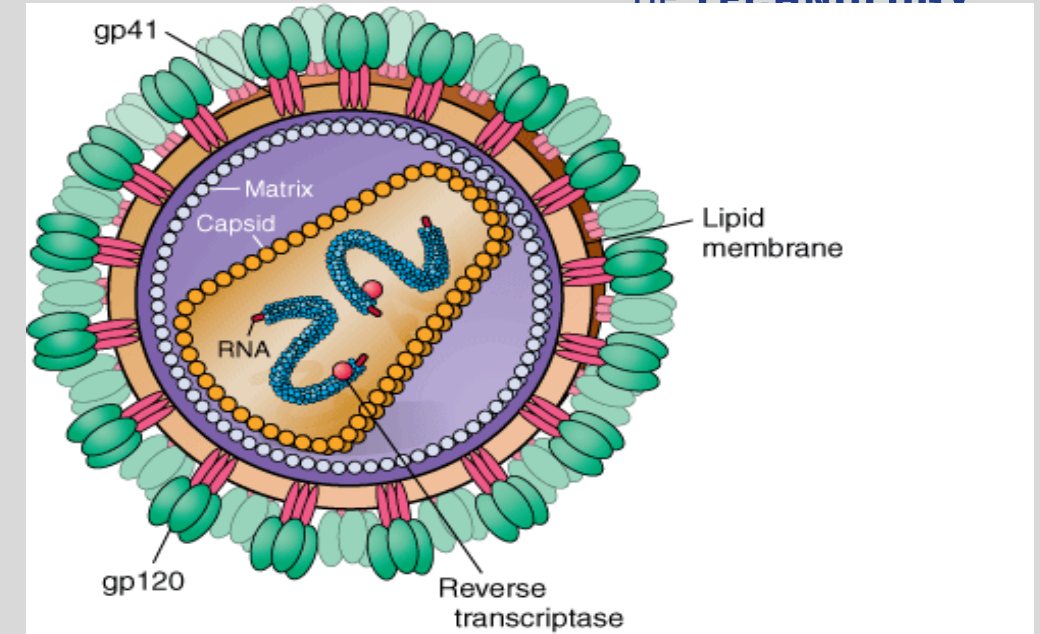
1.American Medical News amednews.com [Christine S. Moyer](#) amednews staff — Posted July 4, 2011

2.Institutes of Medicine Report: [HIV Screening and Access to Care: Health Care System Capacity for Increased HIV Testing and Provision of Care](#) The National Academic Press(2011)

<http://www.iom.edu/Activities/PublicHealth/HIVScreeningCare.aspx>

Human immunodeficiency viruses (HIV)

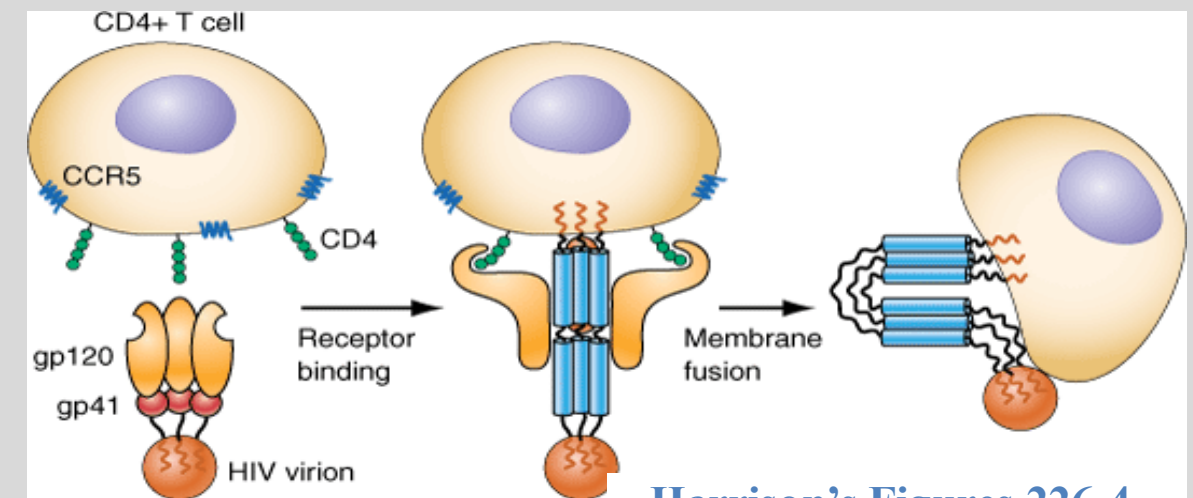
- Nucleocapsid core; lipid bilayer envelope (derived from the host cell plasma membrane)
 - *gag* → 3 structural proteins
 - *pol* → 3 enzymes
 - *env* → large envelope protein
 - several small genes
-
- gp120 binds CD4 → gp120 conformational change → binds to chemokine receptor → exposes gp41 → attachment to host cell → gp41 forms hairpin structure drawing the two membranes together → fusion



B

Harrison's, 19e; Figure 226-2B

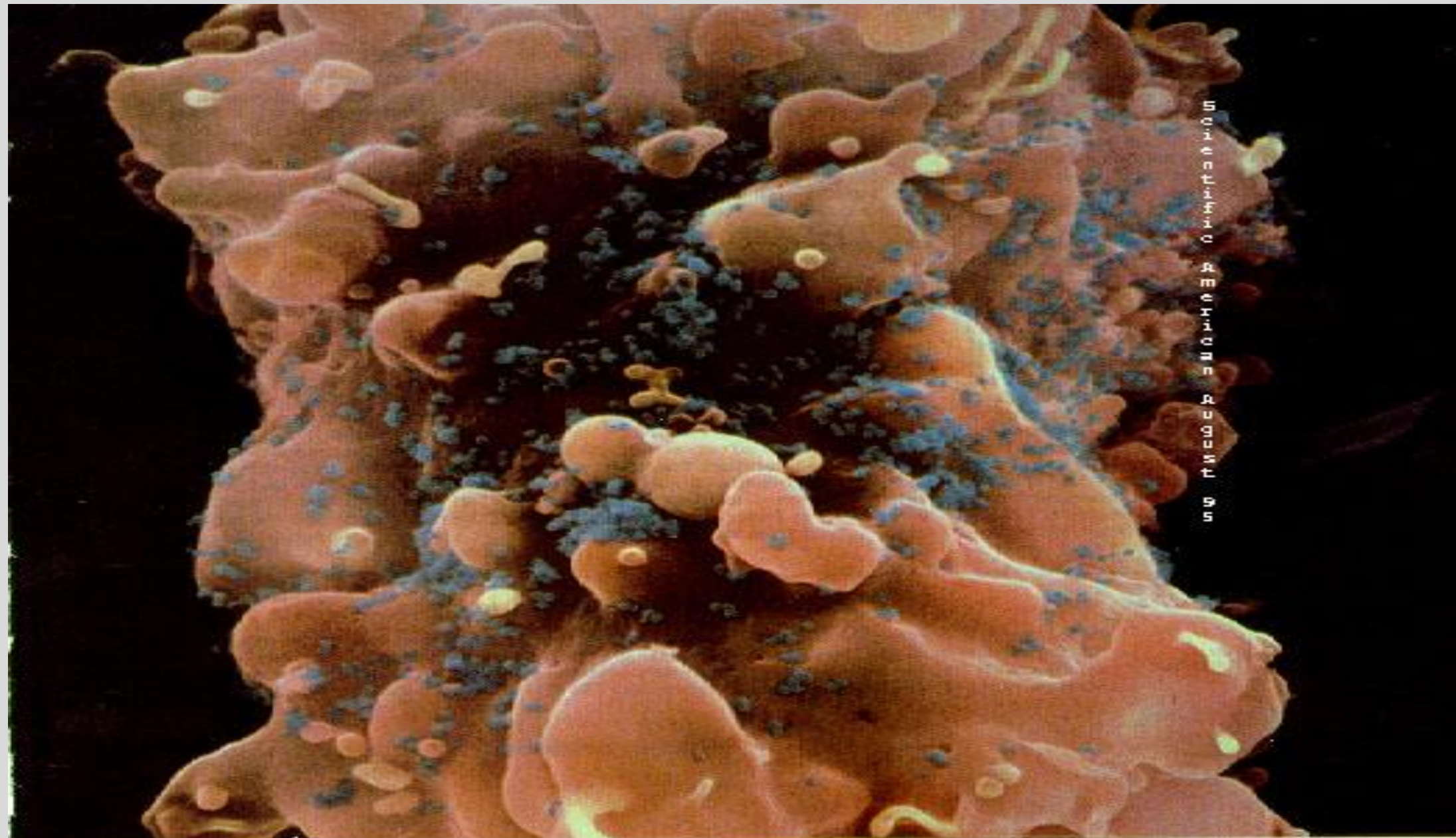
Source: Fauci AS, Kasper DL, Braunwald E, Hauser SL, Longo DL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine*, 17th Edition: <http://www.accessmedicine.com>
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Harrison's Figures 226-4

Source: Fauci AS, Kasper DL, Braunwald E, Hauser SL, Longo DL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine*, 17th Edition: <http://www.accessmedicine.com>
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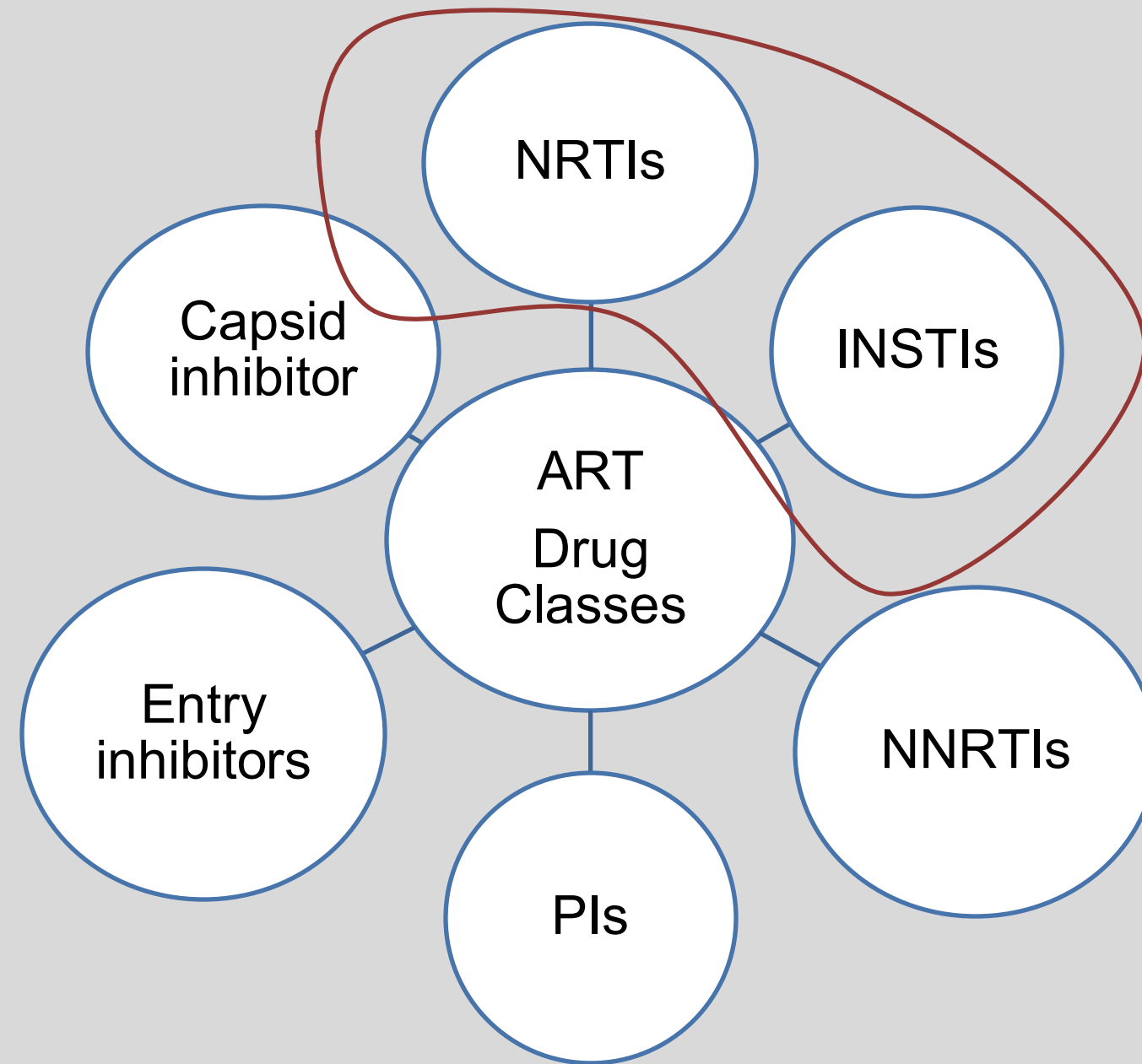
HIV budding from infected white cell



PARTICLES OF HIV (blue spheres), the virus that causes AIDS, bud from an infected white blood cell before moving on to

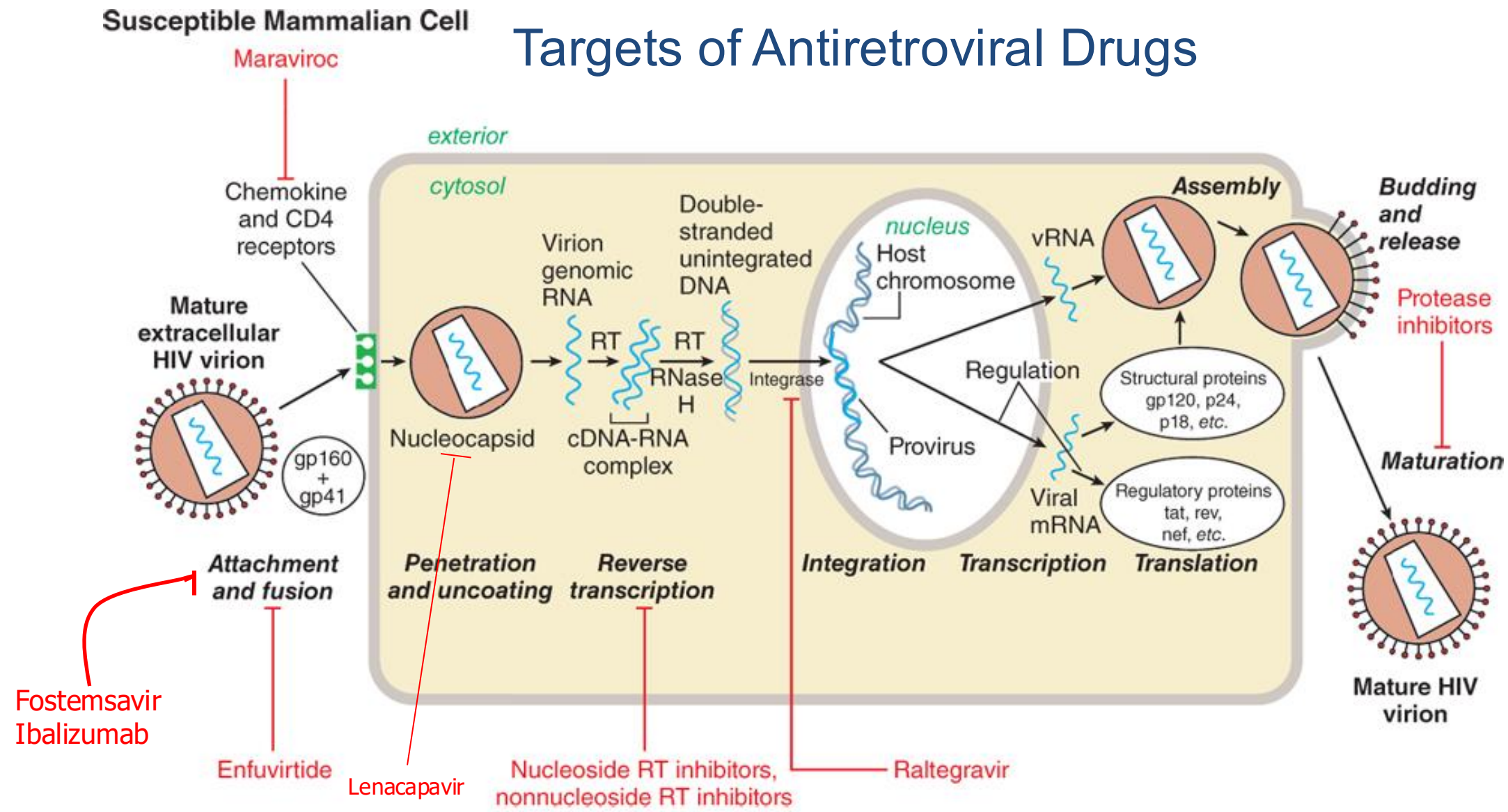
infect other cells. The immune system controls such spread at first but is eventually outmaneuvered by the virus.

Antiretroviral Drug Classes



Many ARV drugs are formulated as 2-drug and 3-drug fixed-dose combinations for improved adherence to therapy.

Here is a YouTube video you may find helpful: <https://www.youtube.com/watch?v=l-2ilZA-aw>



Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.

Replicative cycle of HIV-1 showing the sites of action of available antiretroviral agents. Available antiretroviral agents are shown in red. In this figure, gp120 + gp41 indicates extracellular and intracellular domains, respectively, of envelope glycoprotein.

Viral Resistance to Antiretroviral Drugs

HIV reverse transcriptase is error prone and lacks a proofreading function → mutation is frequent

- HIV has a high mutation rate and a tremendous number of infectious virions.

Transmitted drug resistance is acquired through person to person spread.

Virologic failure:
Inability to achieve or maintain suppression of viral replication to an HIV RNA level <200 copies/mL.

There is a high likelihood that all untreated infected individuals harbor viruses with single amino acid mutations that confer some degree of resistance.

Goals of Antiretroviral Therapy (ART)

1. Maximally and durably suppress plasma HIV RNA (viral load)
2. Restore and preserve immunologic function
3. Reduce HIV-associated morbidity
4. Prolong the duration and quality of survival
5. Prevent HIV transmission

Antiretroviral therapy (ART) is recommended in all HIV-infected individuals to prevent transmission.

🔑 Consistent and effective use of ART resulting in sustained reduction in viral load, in conjunction with:

- consistent condom usage,
- safer sex and drug use practices, and
- detection and treatment of STDs

are essential tools for prevention of sexual and blood-borne transmission of HIV.

Principles of ART

ART has dramatically reduced HIV-associated morbidity and mortality and has transformed HIV infection into a manageable chronic condition, with life expectancy approaching that for people without HIV.

Current standard of care is 3-4 antiretroviral agents for entire duration of treatment.

2 NRTIs + an INSTI, or an NNRTI, or a boosted PI

Preferred initial ARV regimen

Some 2-drug regimens are FDA approved in selected patients.

Outcome of initial therapy should be:

- Undetectable viral load (plasma HIV RNA <20-75 copies/mL) within 24 weeks of starting treatment.
- CD4 count increase of 50-150 cells/mm³ in first year

Treatment failure generally requires implementation of a completely new combination of drugs.

It is usually a consequence of nonadherence.

NRTI: Nucleoside reverse transcriptase inhibitor; NNRTI: Nonnucleoside reverse transcriptase inhibitor;
PI: Protease inhibitor; INST: Integrase strand transfer inhibitor

Adverse Effects of Antiretroviral Therapy

Immune reconstitution inflammatory syndrome:

- reversal of immunodeficiency → ↑ host immune system → inflammatory reaction to overt or subclinical opportunistic infections or malignancies

Class- and drug-specific toxicities:

- Later-generation drugs are generally more tolerable than the first-generation versions.

Significant pharmacokinetic drug interactions:

- substrates, inhibitors and/or inducers of metabolic enzymes and transporters

NRTIs

Nucleoside / Nucleotide

Reverse Transcriptase Inhibitors

- * Abacavir
- * Lamivudine
- * Emtricitabine
- * Tenofovir
- * Zidovudine

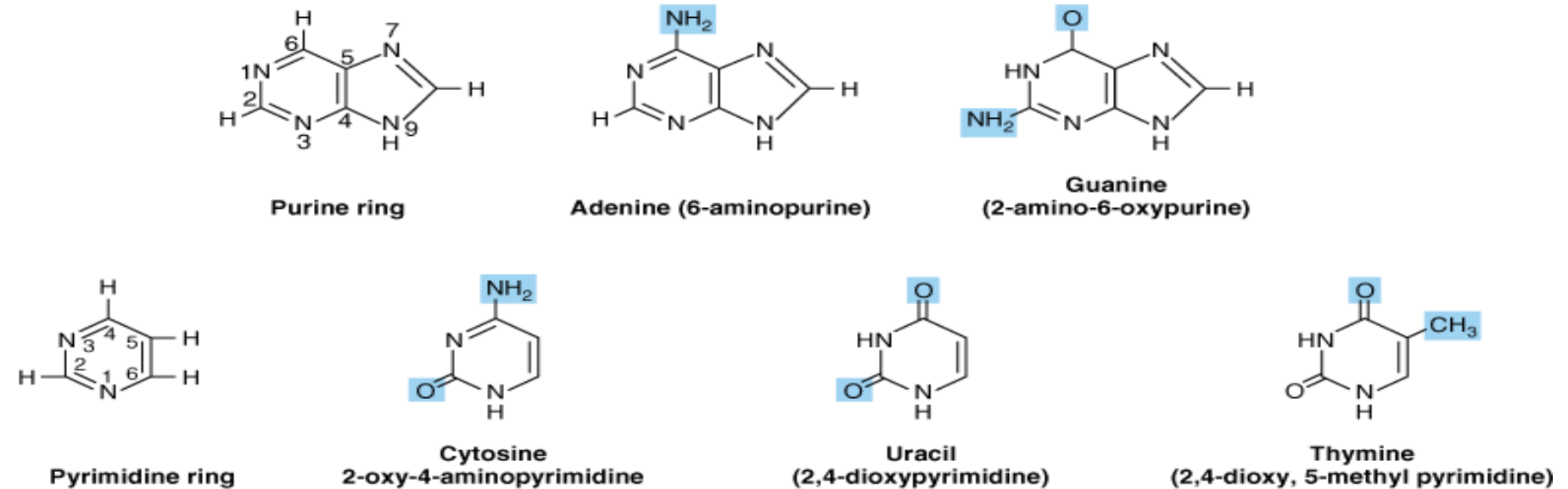
2 NRTIs: The “backbone” of ART regimens

- Activity against HIV-1 and HIV-2.
- Approved for HIV-1 treatment with other ARVs
- Some agents suppress HBV replication, which is an important consideration for HIV/HBV coinfecting patients.

Didanosine, stavudine, and zalcitabine are no longer marketed.

NRTIs are nucleoside analogs, except tenofovir which is a nucleotide analog.

Basic Structure of Nucleosides and Nucleotides



Adenosine-5-triphosphate (ATP)

Adenosine-5-diphosphate (ADP)

Adenosine-5-monophosphate (AMP)

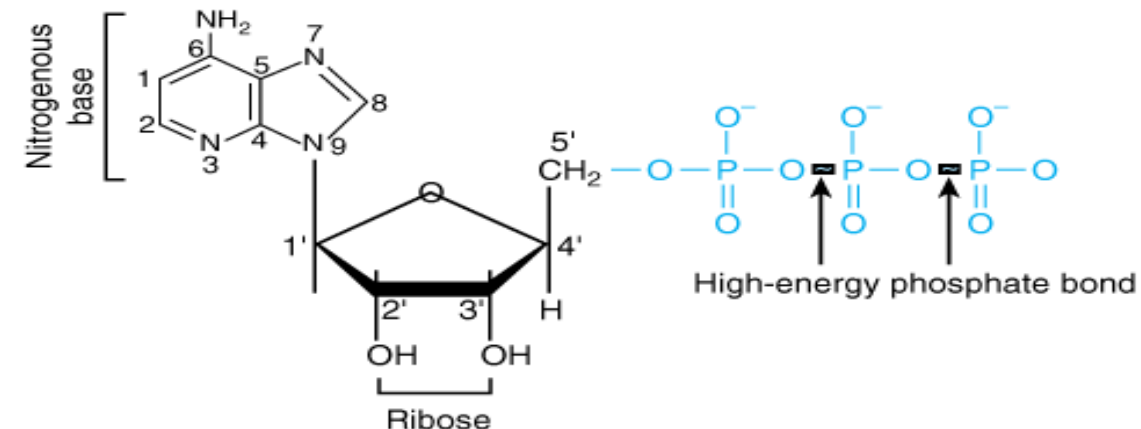
Adenosine

Adenine

Nucleotide

Nucleoside

Purine base

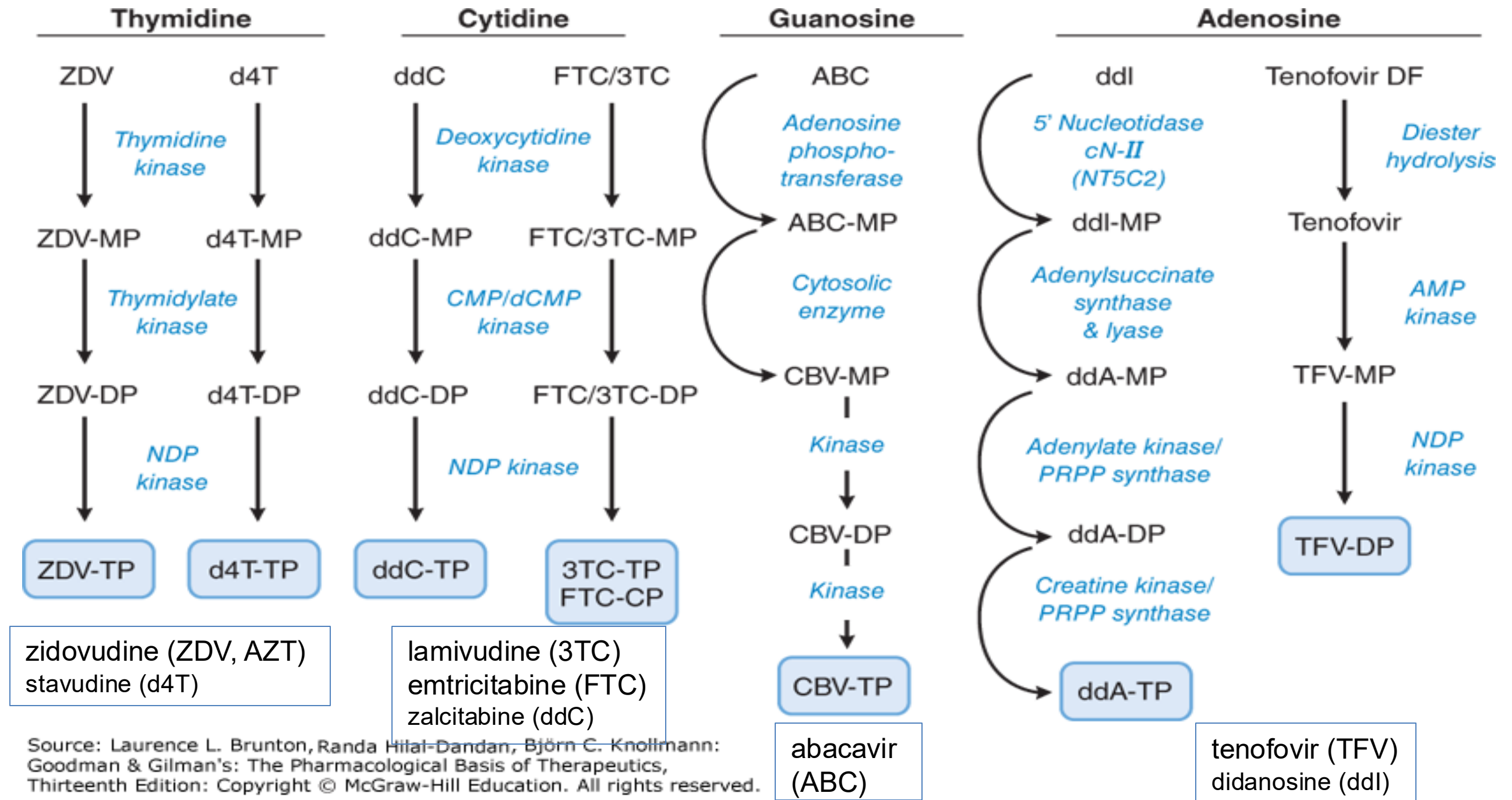


NRTIs lack 3' -OH group

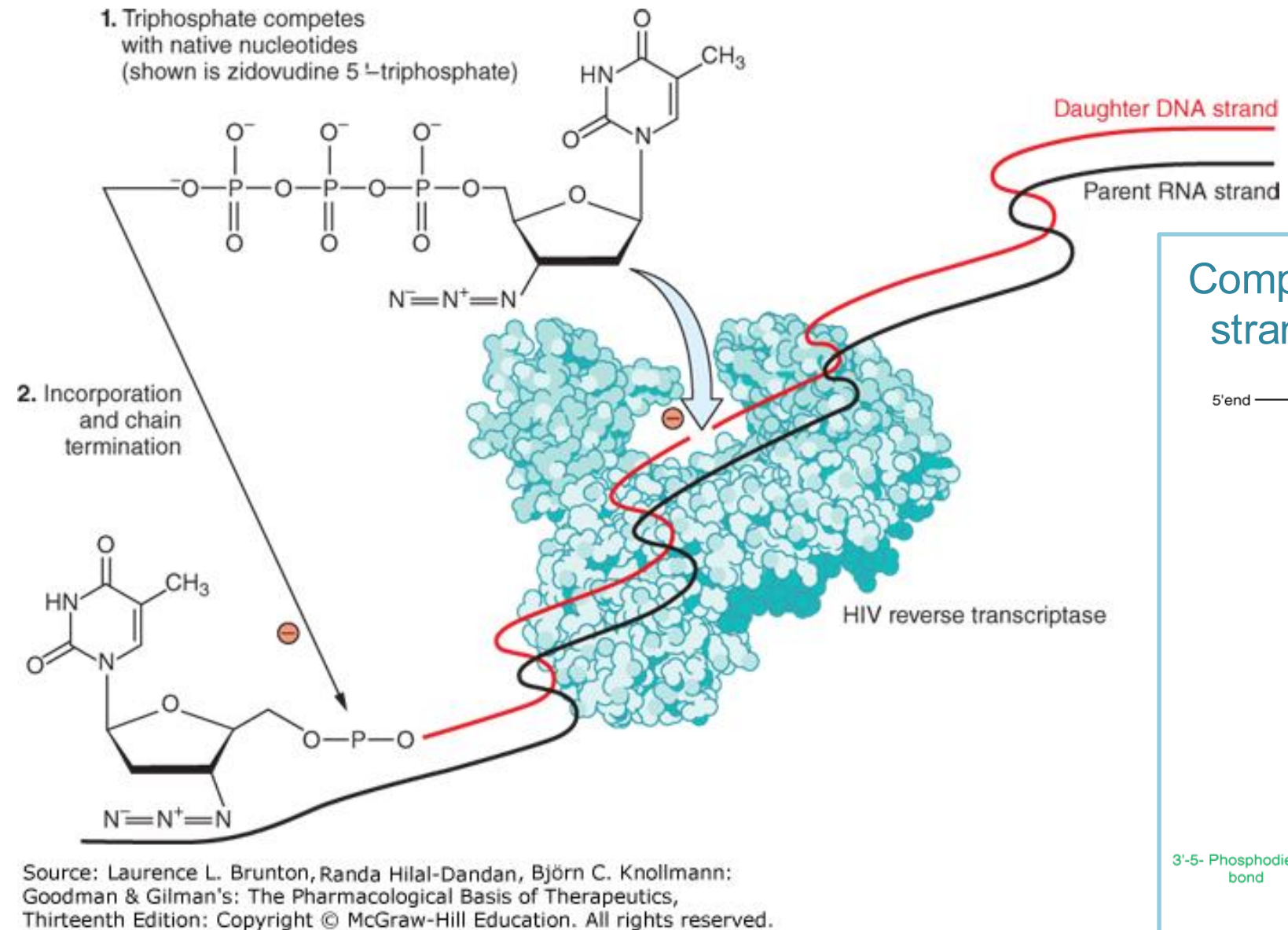
Nucleoside = nucleic acid-ribose

Nucleotide = nucleic acid-ribose-phosphate(s)

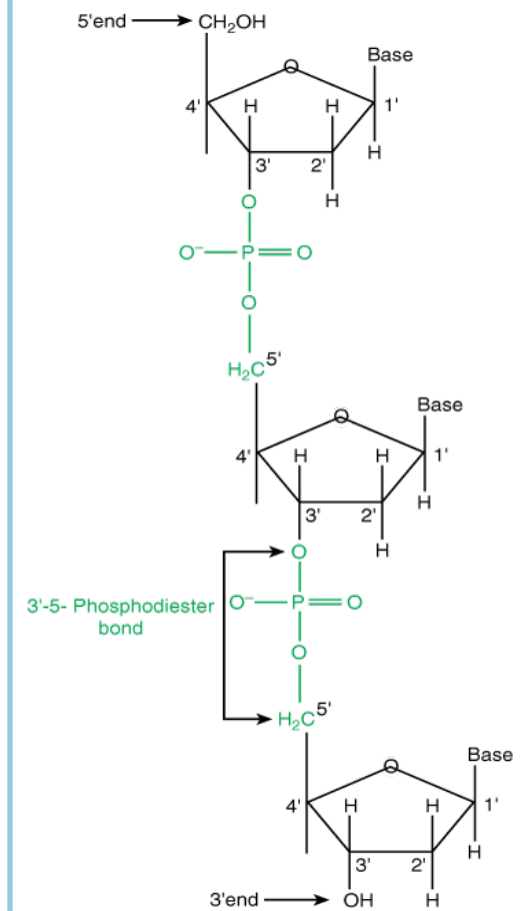
NRTIs are activated by tri-phosphorylation by host kinases



- 1) NRTIs are phosphorylated by cellular enzymes.
- 2) Triphosphorylated drugs competitively inhibit incorporation of native nucleotides into the nascent viral RNA-DNA duplex and
- 3) Terminate elongation of proviral DNA chain due to the absence of the 3' -OH group, which is needed for chain elongation
- 4) Block replication



Compare to normal strand formation:



Source: Janson LW, Tischer ME: *The Big Picture: Medical Biochemistry*: www.accessmedicine.com

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The Big Picture:
Medical Biochemistry
Figure 4-5

Viral Resistance to NRTIs: Point Mutations

- Usually, high level resistance requires 3 to 4 codon substitutions.
- However, **M184V/I** single amino acid substitution
 - Confers high-level resistance to **lamivudine and emtricitabine**
 - Confers low-level resistance to **abacavir**
 - But restores susceptibility to **tenofovir and zidovudine**.
- Other common mutations are K65R and L74R
- Cross-resistance is reported following prolonged therapy.

M=methionine, V=valine, I=isoleucine, K=lysine, R=arginine, L=leucine

M184V/I single mutation of methionine-to-valine or isoleucine substitution at position 184 position.

Inhibiting host DNA polymerase- γ (mitochondrial enzyme)



Mitochondrial Toxicity

Strength of DNA polymerase- γ inhibition:

In vitro studies

Zalcitabine (high toxicity) >> didanosine > stavudine $\geq$ zidovudine >>>

tenofovir = lamivudine = emtricitabine = abacavir

—The second-generation NRTIs have very low mitochondrial toxicity.—

Manifestations

myopathy
peripheral
neuropathy
lipoatrophy
lactic acidosis
hepatic steatosis

Possibly related to
mitochondrial toxicity:
pancreatitis
bone marrow suppression
cardiomyopathy

Hepatic steatosis and hepatic failure with lactic acidosis occurs rarely and can be fatal.

🔔 NRTI treatment should be suspended in the setting of rapidly rising aminotransferase levels, progressive hepatomegaly, or metabolic acidosis of unknown cause.

Features of Specific NRTIs: Tenofovir DF (TDF) and Tenofovir AF (TAF)

Adenosine analogs.

Oral, well absorbed, P-glycoprotein substrate, minimal hepatic metabolism, urine as tenofovir, $t_{1/2\text{intracellular}}$ 10-50 h

TDF is rapidly converted after absorption by hydrolases to tenofovir, which is taken up by host cells and metabolized to TFV-DF.

TAF is administered at a lower dose than TDF → lower plasma concentrations.

However, intracellular concentrations of active tenofovir diphosphate are higher.

TAF circulates as uncleaved prodrug, which is taken up into cells. TAF is then metabolized intracellularly to TFV-DF.

TDF and TAF have broad therapeutic applications and are available in numerous fixed-dose combinations.

- First-line NRTIs for treatment of HIV-1
- Pregnant patients, all trimesters
- TDF Children ≥ 2 years old
- TAF Children, adolescents ≥ 12 years old
- Pre-exposure prophylaxis (PrEP)
- Post-exposure prophylaxis (PEP)
- Chronic **HBV** treatment and prophylaxis in immunocompromised patients

Both TDF and TAF suppress HBV infection → Severe acute exacerbation of HBV infection and hepatitis can occur in HIV/HBV coinfecting patients treated with TDF or TAF if the NRTI is discontinued. (US Boxed Warning)

TDF and TAF Adverse Effects and Drug Interactions

TDF and TAF are generally well tolerated.

- **TAF** has lower plasma concentrations and has less long-term kidney and bone toxicity compared with TDF.

Main toxicities: Kidney and bone adverse effects

- **Renal impairment:** Increases in serum creatinine, proteinuria, glycosuria, hypophosphatemia, acute tubular necrosis
- **Kidney injury:** Proximal tubular dysfunction (Fanconi syndrome), acute kidney injury, chronic kidney disease, nephrogenic diabetes insipidus
 - Monitor renal function. Avoid TDF (if possible) in patients with baseline risk factors for kidney injury.
- **Bone loss:** Decreased bone mineral density

Drug interactions:

- Tenofovir + abacavir → high rates of virologic failure
- P-glycoprotein inhibitors: Ritonavir or cobicistat boosted drugs may increase tenofovir levels.
- TAF only – P-gp inducers are contraindicated: Rifampin, carbamazepine, oxcarbazepine, phenytoin, phenobarbital ↓TAF levels → ↓ efficacy

NRTIs Drug-specific properties: Lamivudine and Emtricitabine

Cytidine analogs. Emtricitabine is fluorinated lamivudine. *They should not be used together.*

Oral, well absorbed	$t_{1/2}^{\text{intracellular}}$
Good bioavailability	lamivudine 10-15 h (high dose for 1x daily dosing),
excreted in urine as unchanged drug	emtricitabine ~40 h (once daily)

- Both drugs suppress HBV infection
- Severe acute exacerbation of HBV infection and hepatitis can occur in HIV/HBV coinfectd patients treated with lamivudine or emtricitabine if the NRTI is discontinued. (US Boxed Warning)

- Resistance: M184V/I point mutations → high level resistance.
- K65R, L74R also reduce susceptibility.

Well tolerated

Emtricitabine may cause hyperpigmentation, usually on the hands and/or soles of the feet

Features of Specific NRTIs: Abacavir and Zidovudine

Abacavir guanosine analog

Oral, well absorbed, hepatic metabolism by alcohol dehydrogenase, glucuronyl transferase, urine as metabolites, $t_{1/2\text{intracellular}}$ 12-26 h
Intracellular metab.: abacavir monophosphate → carbovir MP →→ active carbovir TP

Pharmacogenomics: All patients must be screened for HLA-B*5701

- Hypersensitivity reactions: multiorgan involvement, *potentially fatal*

Abacavir should not be given to patients with the HLA-B*5701 genotype nor restarted in patients who have a hypersensitivity reaction.

- Resistance: M184V/I and K65R frequently found in patients receiving abacavir
- Frequently paired with lamivudine as a component of NRTI backbone
- Avoid using with tenofovir due to high rate of virologic failure with the combination
- Alcohol (EtOH) → ↑ abacavir levels

Zidovudine thymidine analog

Oral (2x daily), IV (continuous infusion), hepatic glucuronidation, urine unchanged drug, $t_{1/2\text{intracellular}}$ ~7 h

Therapeutic uses:

- Perinatal IV infusion to prevent vertical transmission
- ART for neonates

Zidovudine is otherwise rarely used: twice daily dosing, more severe adverse effects, and inferior virologic potency (zidovudine-lamivudine v. TDF-emtricitabine ART)

Questions

1. What are the important (clinically relevant) take-aways about the PK properties of the NRTIs class?
2. How is viral resistance related to NRTIs' mechanism of action and effects?
3. How is mitochondrial toxicity related to the NRTIs' mechanism of action?

Questions

1. What are the main toxicities of tenofovir and abacavir.
2. Which of the second-generation NRTIs should not be combined and why?
3. What screening test must be performed prior to administering abacavir and why?
4. What drugs can cause reactivation of HBV and hepatitis when withdrawn in patients with HIV/HBV coinfection?

INSTIs

HIV Integrase Strand Transfer Inhibitors

- *Bictegravir (BIC)
- *Dolutegravir (DTG)
- *Elvitegravir (ELV)
- *Raltegravir (RAL)
- *Cabotegravir (CAB)

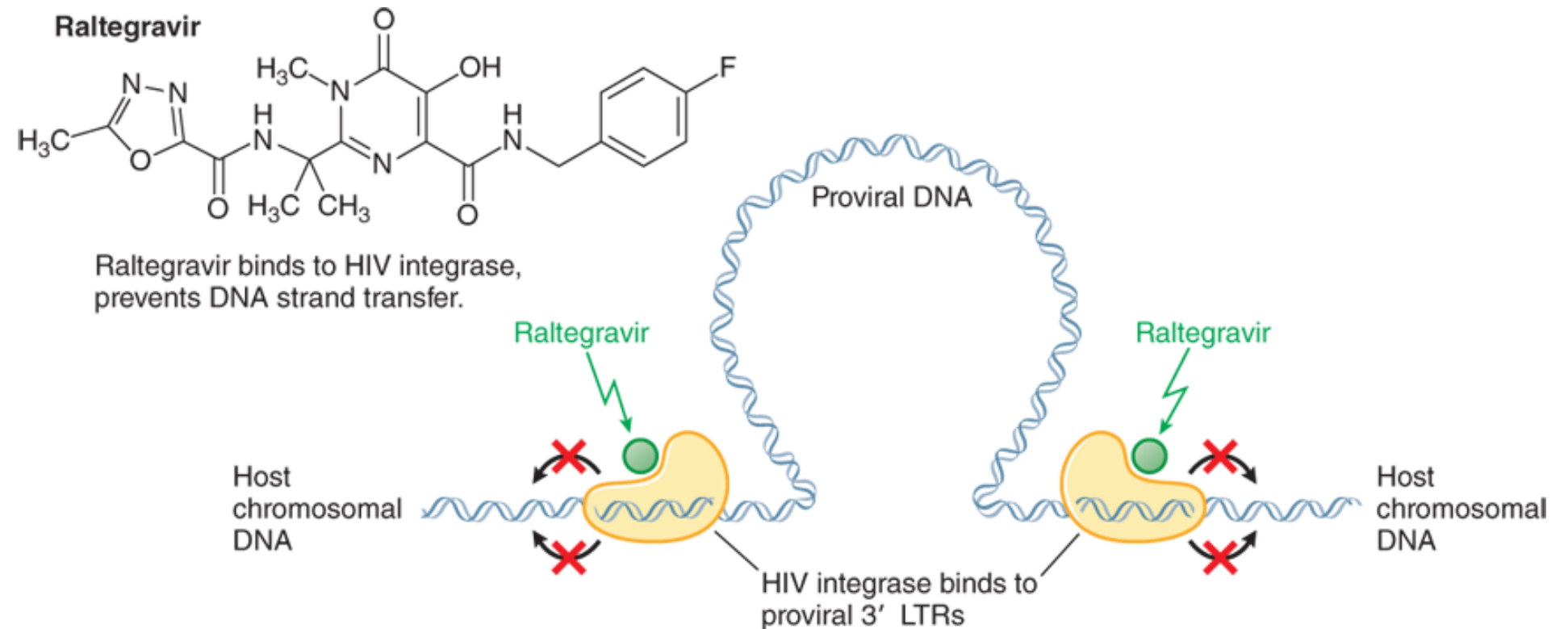
Activity against HIV-1 and HIV-2

INSTIs are well tolerated, highly effective, and have a high barrier to resistance.

INSTI Characteristics

Characteristics	Second-generation INSTIs		First-generation INSTIs	
	BIC	DTG	ELV	RAL
Dosing	Once daily	Once daily 2x/d if used with CYP3A4 or UGT1A1 inducers	Once daily Cobicistat boost required.	400 mg 2x daily 1200 mg 1x daily
Single drug tab Fixed-dose combos	no BIC-TAF-emtricitabine	yes DTG-ABC-emtricitabine DTG-lamivudine	no ELVc-TDF-emtricitabine ELVc-TAF-emtricitabine	yes no
Virologic efficacy: ELV or RAL Resistant HIV	In vitro: yes Clinical trial data are not available.	Yes, some isolates Higher dose 2x/d	no	no
Adverse effects	Weight gain, nausea, diarrhea, headache, insomnia Depression and suicidality are rare, occurring mainly in patients with preexisting psychiatric condition.			
	↑CK (4%)	Hypersensitivity Hepatotoxicity ↑CK, myositis	↑triglycerides ↑LDL cholesterol	↑CK, myopathy Hypersensitivity SJS/TEN
CYP3A4 DDIs	CYP3A4 substrate	CYP3A4 substrate (minor)	CYP3A4 substrate COBI 3A4 inhibitor	No
Antacids Supplements	Oral absorption of all INSTIs may be reduced by polyvalent cations, such as calcium, magnesium, aluminum.			
Other significant DDIs	P-gp and UGT1A1 substrate	P-gp and UGT1A1 substrate	UGT1A1 substrate COBI Pgp inhibitor	UGT1A1 substrate

- INSTIs block the catalytic activity of the HIV-encoded integrase → prevents integration of virus DNA into the host chromosome
- Humans lack integrase, and human DNA is not known to undergo excision and reintegration.



Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.

Mechanism of action of the HIV integrase inhibitor raltegravir

INSTIs Therapeutic Uses and Resistance

Activity (as a class)

- Potent against both HIV-1 and HIV-2
- Treatment-naïve and treatment-experienced
 - Adults
 - Adolescents, and
 - Pediatrics

The INSTIs are generally well tolerated.

Headache and GI effects are the most frequently reported side effects.

Cabotegravir: Injection site reactions

Resistance

- Primary mutations in integrase gene (Q184R/H/K, or N155H) and
- Additional secondary mutations can accumulate and cause high-level resistance and cross-resistance.

Dolutegravir and bictegravir have a high barrier to resistance.

Raltegravir and elvitegravir have a lower barrier to resistance.

Cabotegravir ART: long-acting IM suspension

Cabotegravir is available in oral form, as a long-acting IM suspension for injection and fixed-dose cabotegravir and rilpivirine long-acting suspension for injection.

For gluteal administration only (may not be an option for people with buttocks implants or fillers).

1. Treatment: Cabotegravir + rilpivirine

- Oral lead-in (optional, to determine tolerability):
 - daily oral cabotegravir + daily oral rilpivirine x 1 month
 - Continuation: IM cabotegravir-rilpivirine every 4 to 8 weeks (dosing differs)
- Oral bridging therapy: Patients who plan to miss a scheduled dose of IM CAB-RPV by >7 days.

Required criteria: HIV-1 infections in adults and adolescents...

Who have no baseline resistance to either medication,

Who have no prior virologic failures,

Who do not have active hepatitis B virus (HBV) infection (unless also receiving an oral HBV active regimen),

Who are not pregnant and are not planning on becoming pregnant, *and*

Who are not receiving medications with significant drug interactions with oral or injectable CAB or RPV.

2. PrEP: HIV-negative individuals, IM cabotegravir with or without oral lead-in

Questions

1. What properties make INSTIs the preferred first-line choice for ART backbone?
2. Which two are the first-line INSTIs?
3. What is the role of cabotegravir-rilpiverine 2-drug ART in the treatment of HIV-1 patients?
4. What is another therapeutic use of cabotegravir IM?

Summary of NRTIs and NNRTIs

- Antiretroviral therapy (ART) consists of a combination of antiretroviral drugs that suppress HIV replication. The combination increases potency against the virus and reduces the likelihood of the virus developing resistance. ART reduces the morbidity and mortality rates and improves quality of life. Suppressing viral replication below a detectable level reduces transmission. ART is recommended for everyone who has HIV. Adherence to therapy is essential for suppression of replication. The patient's barriers to adherence should be assessed prior to therapy and at every visit.
- Drugs targeting every step in the HIV replication cycle have been developed. The four classes of drugs for initial treatment are NRTIs, INSTIs, NNRTIs, and PIs.
- Resistance arises from point mutations in binding sites. Transmitted drug resistance is acquired through person to person spread. There is a high likelihood that all untreated infected individuals harbor viruses with single amino acid mutations that confer some degree of resistance.

- Two second-generation NRTIs form the ART backbone for initial therapy: tenofovir disoproxil fumarate, tenofovir alafenamide, emtricitabine, lamivudine, and abacavir. Lamivudine and emtricitabine are structurally similar and not used together. Tenofovir and abacavir should not be used together because of high rates of virologic failure.
- NRTIs have a low affinity for host DNA polymerase except mitochondrial DNA polymerase-gamma.
- Many NRTI toxicities are believed to be due to mitochondrial toxicity: myopathy, peripheral neuropathy, pancreatitis, lipoatrophy, and hepatic steatosis, and lactic acidosis, which can be fatal. Pancreatitis, bone marrow suppression, and cardiomyopathy may be related to mitochondrial toxicity. First-generation NRTIs are most associated with mitochondrial toxicities. Tenofovir, TDF, can cause proximal renal tubulopathy and bone mineral loss. TAF is administered in lower doses and has less kidney and bone toxicity than TDF. Abacavir can cause life-threatening hypersensitivity reaction. Screening for HLA-B*5701 allele is necessary prior to starting therapy.
- NRTIs are oral and excreted in the urine. Abacavir is metabolized by alcohol dehydrogenase and glucuronyl transferase.

- Integrase strand transfer inhibitors (INSTIs) are bictegravir, dolutegravir, elvitegravir (boosted with cobicistat), raltegravir, and cabotegravir. BIC and DOL are first-line ART backbone with 2 NRTIs for treatment-naïve patients. Cabotegravir long-acting IM suspension is a PrEP option and is formulated with rilpivirine as 2-drug ART for treatment of selected, stable HIV-1 patients. CAB is formulated as a tablet for oral lead-in or bridging therapy.
- INSTIs inhibit the insertion of proviral DNA into the host cell genome.
- Resistance occurs with single point mutations in the integrase gene. BIC and DOL have a high barrier to resistance. RAL and ELV have a lower barrier to resistance.
- The INSTIs are taken orally once daily (raltegravir twice daily or once daily at a higher dose). DOL, RAL, and CAB are UGT1A1 substrates. BIC and ELV are CYP3A4 substrates. Efficacy can be reduced when administered with UGT1A1, CYP3A4, and P-glycoprotein inducers.
- The INSTIs are well tolerated. Headache and GI discomfort are the most frequently reported side effects.

Living with HIV: Steps to Better Health



Be active every day

M T W T F S S



Take ART (antiretroviral treatment)
as directed



Keep all
appointments with
your health care
providers



Eat healthy food
that is safely
prepared

Staying Healthy With HIV



Get recommended vaccinations



Don't smoke or use drugs



Limit alcohol



Use condoms during sex

For more information, visit HIVinfo.NIH.gov.

References

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